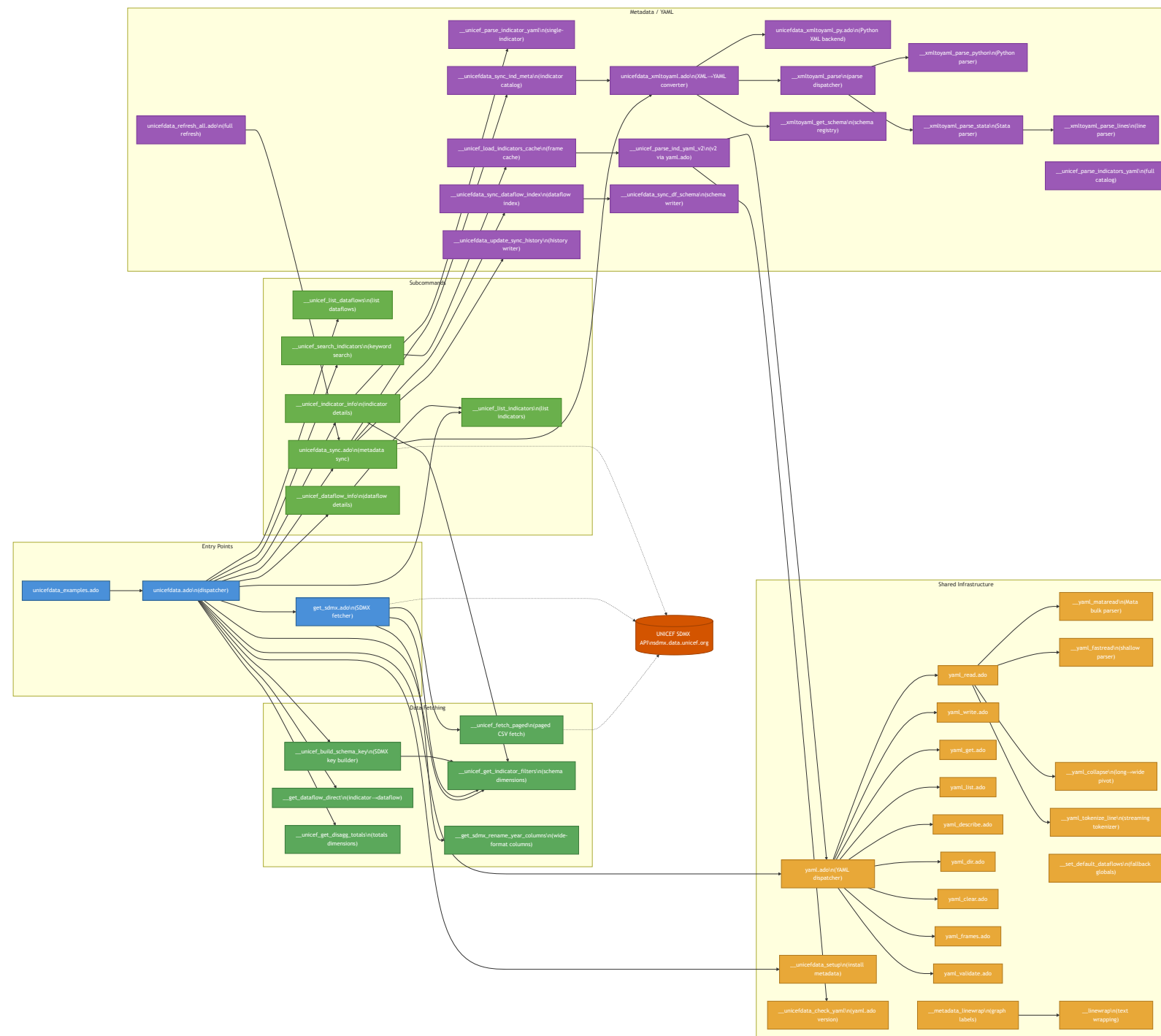


unicefData Stata Package — Module Architecture

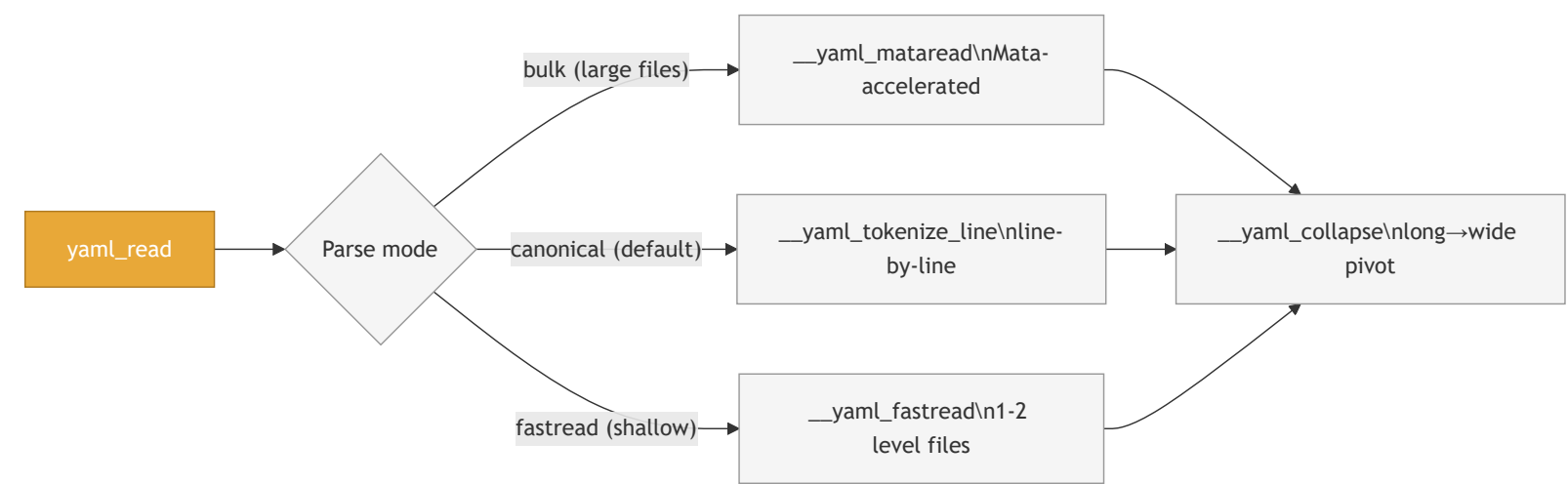
Module Dependency Diagram



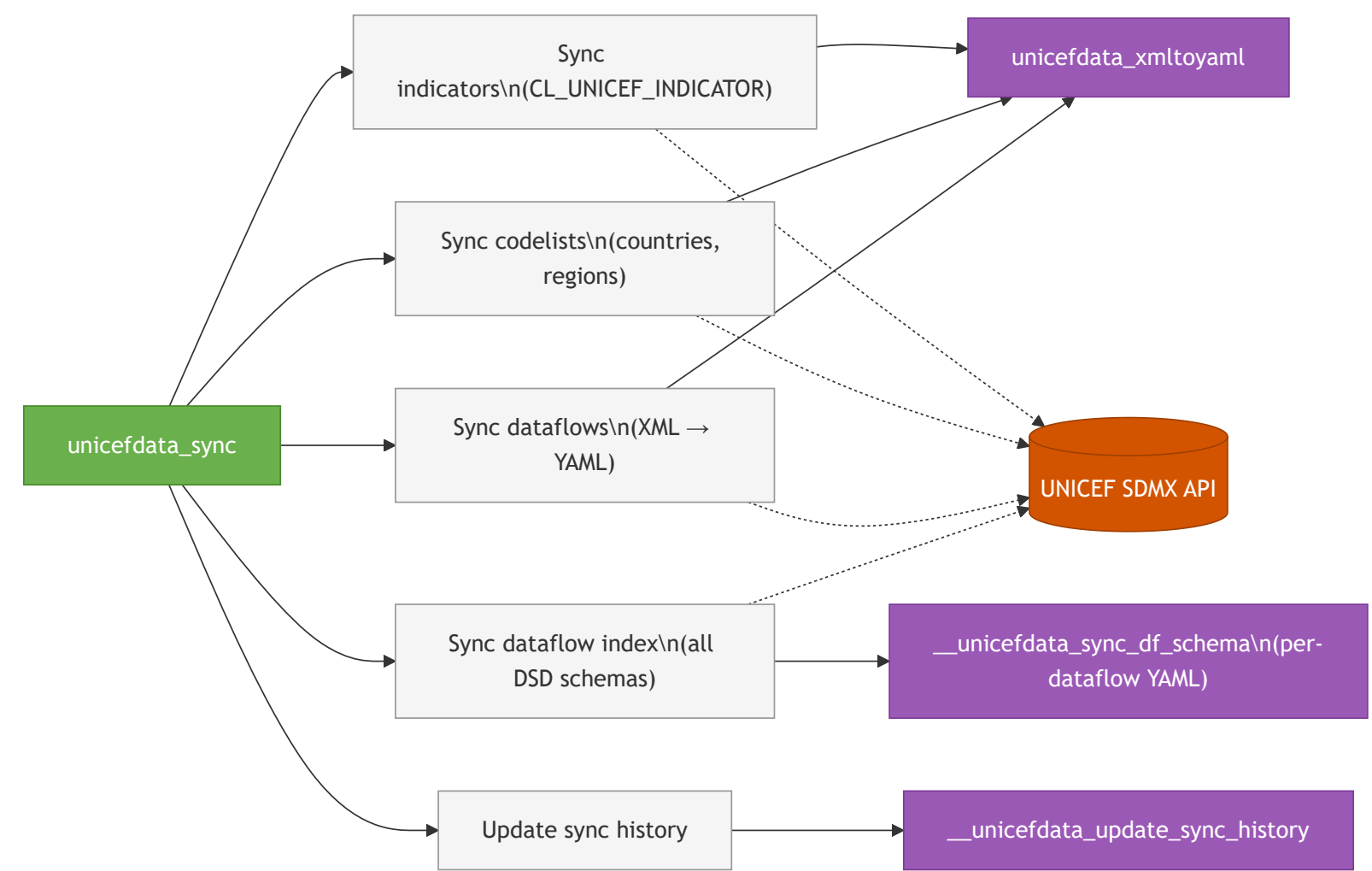
XML→YAML Parser Selection (unicefdata_xmltoyaml.ado)



YAML Read Paths (`yaml_read.ado`)



Metadata Sync Pipeline (`unicefdata_sync.ado`)



Dataflow Resolution (unicefdata.ado)



Module Summary

Module	Role	Layer	Calls
unicefdata.ado	Main entry point & dispatcher	Entry	get_sdmx , __unicef_list_dataflows , __unicef_search_indicators , __unicef_list_indicators , __unicef_indicator_info , __unicef_dataflow_info , unicefdata_sync , __get_dataflow_direct , __unicef_get_disagg_totals , __unicef_build_schema_key , __unicef_get_indicator_filters , __unicefdata_setup , yaml
get_sdmx.ado	Low-level SDMX data fetcher	Entry	__unicef_fetch_paged , __unicef_get_indicator_filters , __get_sdmx_rename_year_columns
unicefdata_examples.ado	Interactive examples (12)	Entry	unicefdata
unicefdata_sync.ado	Metadata sync orchestrator	Subcommand	unicefdata_xmltoyaml , __unicefdata_sync_dataflow_index , __unicefdata_sync_ind_meta , __unicefdata_update_sync_history
unicefdata_refresh_all.ado	Full refresh orchestrator	Subcommand	unicefdata_sync , Python scripts
__unicef_list_dataflows.ado	List available dataflows	Subcommand	(leaf)

Module	Role	Layer	Calls
<code>__unicef_list_indicators.ado</code>	List indicators for a dataflow	Subcommand	(leaf)
<code>__unicef_search_indicators.ado</code>	Keyword search over indicators	Subcommand	<code>__unicef_load_indicators_cache</code>
<code>__unicef_indicator_info.ado</code>	Display indicator metadata	Subcommand	<code>__unicef_parse_indicator_yaml</code> , <code>__unicef_get_indicator_filters</code>
<code>__unicef_dataflow_info.ado</code>	Display dataflow schema details	Subcommand	<code>__unicef_list_indicators</code>
<code>get_sdmx.ado</code>	SDMX REST API client	Fetch	<code>__unicef_fetch_paged</code> , <code>__unicef_get_indicator_filters</code> , <code>__get_sdmx_rename_year_columns</code>
<code>__unicef_fetch_paged.ado</code>	Paged CSV download from API	Fetch	(leaf — calls API directly)
<code>__unicef_build_schema_key.ado</code>	Build SDMX filter key	Fetch	<code>__unicef_get_indicator_filters</code>
<code>__unicef_get_indicator_filters.ado</code>	Read schema dimensions from YAML	Fetch	(leaf)
<code>__get_dataflow_direct.ado</code>	Indicator → dataflow YAML lookup	Fetch	(leaf)
<code>__unicef_get_disagg_totals.ado</code>	Get dimensions with _T totals	Fetch	(leaf)
<code>__get_sdmx_rename_year_columns.ado</code>	Rename year columns (wide format)	Fetch	(leaf)

Module	Role	Layer	Calls
<code>unicefdata_xmltoyaml.ado</code>	XML→YAML converter (dispatcher)	Metadata	<code>__xmltoyaml_get_schema</code> , <code>unicefdata_xmltoyaml_py</code> , <code>__xmltoyaml_parse</code>
<code>unicefdata_xmltoyaml_py.ado</code>	Python XML backend	Metadata	Python script <code>unicefdata_xml2yaml.py</code>
<code>__xmltoyaml_parse.ado</code>	Parse dispatcher (Python→Stata)	Metadata	<code>__xmltoyaml_parse_python</code> , <code>__xmltoyaml_parse_stata</code>
<code>__xmltoyaml_parse_stata.ado</code>	Stata XML parser	Metadata	<code>__xmltoyaml_parse_lines</code>
<code>__xmltoyaml_parse_lines.ado</code>	Line-by-line XML element parser	Metadata	(leaf)
<code>__xmltoyaml_parse_python.ado</code>	Python XML parser wrapper	Metadata	Python script <code>unicefdata_xml2yaml.py</code>
<code>__xmltoyaml_get_schema.ado</code>	SDMX type schema registry	Metadata	(leaf)
<code>__unicef_parse_indicator_yaml.ado</code>	Single- indicator YAML parser	Metadata	(leaf)
<code>__unicef_parse_indicators_yaml.ado</code>	Full catalog YAML parser (v1)	Metadata	(leaf)
<code>__unicef_parse_ind_yaml_v2.ado</code>	Full catalog YAML parser (v2)	Metadata	<code>__unicefdata_check_yaml</code> , <code>yaml</code>
<code>__unicef_load_indicators_cache.ado</code>	Frame-cached indicator loader	Metadata	<code>__unicef_parse_ind_yaml_v2</code>

Module	Role	Layer	Calls
<code>__unicefdata_sync_dataflow_index.ado</code>	Sync all dataflow schemas	Metadata	<code>__unicefdata_sync_df_schema</code>
<code>__unicefdata_sync_df_schema.ado</code>	Write single dataflow schema YAML	Metadata	(leaf)
<code>__unicefdata_sync_ind_meta.ado</code>	Sync indicator catalog	Metadata	<code>unicefdata_xmltoyaml</code>
<code>__unicefdata_update_sync_history.ado</code>	Write sync history YAML	Metadata	(leaf)
<code>yaml.ado</code>	YAML subcommand dispatcher	Infrastructure	<code>yaml_read</code> , <code>yaml_write</code> , <code>yaml_get</code> , <code>yaml_list</code> , <code>yaml_describe</code> , <code>yaml_dir</code> , <code>yaml_clear</code> , <code>yaml_frames</code> , <code>yaml_validate</code>
<code>yaml_read.ado</code>	YAML file reader (3 parse paths)	Infrastructure	<code>__yaml_mataread</code> , <code>__yaml_fastread</code> , <code>__yaml_collapse</code> , <code>__yaml_tokenize_line</code>
<code>yaml_write.ado</code>	YAML file writer	Infrastructure	(leaf)
<code>yaml_get.ado</code>	Retrieve YAML key values	Infrastructure	(leaf)
<code>yaml_list.ado</code>	List YAML keys/values	Infrastructure	(leaf)
<code>yaml_describe.ado</code>	Show YAML structure	Infrastructure	(leaf)
<code>yaml_dir.ado</code>	List YAML data in memory	Infrastructure	(leaf)
<code>yaml_clear.ado</code>	Clear YAML data from memory	Infrastructure	(leaf)

Module	Role	Layer	Calls
<code>yaml_frames.ado</code>	List YAML frames	Infrastructure	(leaf)
<code>yaml_validate.ado</code>	Validate YAML against schema	Infrastructure	(leaf)
<code>__yaml_mataread.ado</code>	Mata-accelerated bulk parser	Infrastructure	(leaf)
<code>__yaml_fastread.ado</code>	Shallow YAML fast parser	Infrastructure	(leaf)
<code>__yaml_collapse.ado</code>	Long→wide YAML pivot	Infrastructure	(leaf)
<code>__yaml_tokenize_line.ado</code>	YAML line tokenizer	Infrastructure	(leaf)
<code>__unicefdata_setup.ado</code>	First-run metadata installer	Infrastructure	(leaf)
<code>__unicefdata_check_yaml.ado</code>	Check/install yaml.ado version	Infrastructure	(leaf)
<code>__set_default_dataflows.ado</code>	Hardcoded fallback dataflows	Infrastructure	(leaf)
<code>__linewrap.ado</code>	Text line wrapping	Infrastructure	(leaf)
<code>__metadata_linewrap.ado</code>	Graph-label line wrapping	Infrastructure	<code>__linewrap</code>

Call Flow Example

User: unicefdata, indicator(MNCH_SAB) country(BRA;IND) year(2010/2020)

unicefdata.ado

- └ __unicefdata_setup (ensure YAML metadata installed)
- └ parse subcommand → data retrieval mode
- └ __get_dataflow_direct (look up MNCH_SAB → MNCH dataflow)
 - └ reads _unicefdata_indicators_metadata.yaml
- └ __unicef_get_disagg_totals (SEX has _T → auto-filter SEX=_T)
- └ __unicef_get_indicator_filters (read MNCH schema dimensions)
- └ __unicef_build_schema_key (build key: MNCH.MNCH_SAB.BRA+IND._T...)
 - └ __unicef_get_indicator_filters
- └ get_sdmx (fetch data from SDMX API)
 - └ __unicef_get_indicator_filters (validate dimensions)
 - └ build URL: sdmx.data.unicef.org/ws/public/sdmxapi/rest/data/MNCH/...
 - └ __unicef_fetch_paged (page 1: import CSV)
 - └ HTTP GET → API (100K rows/page)
 - └ page 2... (if needed)
 - └ return to unicefdata.ado

unicefdata.ado

- └ label variables, apply value labels
- └ store char _dta[unicef_*] metadata
- └ return r(N), r(indicator), r(dataflow)